

Week 1

Introduction to Federated Learning

Theory and Applications of Federated Learning

Graduate course · English instruction

Faranak Solat

Department of AI & Data Science

Sejong University



Course Positioning

What this graduate course is designed to achieve

Course focus

Theory, algorithms, system constraints, privacy/security, and real-world applications of federated learning.

Learning mode

Lecture + paper reading + algorithm discussion + application-oriented research thinking.

Semester structure

Foundations

Algorithms

Systems

Privacy & Security

Applications

Research discussion

Week 1 role

Build the conceptual foundation: why FL exists, what makes it different, and where the course is going.

Textbook and References

Two core references for the course

Main Reference

H. Ludwig and N. Baracaldo, Eds., Federated Learning: A Comprehensive Overview of Methods and Applications. Springer, 2022.

Use: main textbook for methods, systems, privacy/security, and applications.

Main book = structured textbook

Supplementary Reference

P. Kairouz, H. B. McMahan, et al., "Advances and Open Problems in Federated Learning," Foundations and Trends in Machine Learning, vol. 14, no. 1–2, pp. 1–210, 2021.

Use: research-oriented survey and open problems.

Survey = research map and open challenges

Class Rules and Expectations

Clear rules help make the course useful for everyone

Language

The class will be conducted in English. Students are encouraged to ask questions in English during class.

Participation

Active participation is expected: questions, short discussions, and paper-based comments are part of the class culture.

Academic integrity

Assignments, reviews, and presentations must properly cite sources. Plagiarism and copied work are not accepted.

Preparation

Students should read assigned materials before class when a paper or chapter is announced.

Communication

Important announcements, materials, and deadlines should be checked regularly through the official class channel.

Late policy

Late submissions may receive a penalty unless there is a valid and approved reason.

Assessment Plan

Tentative grading structure and major evaluation points

Major evaluations

Midterm Exam	Week 8	Weeks 1–7
Final Exam	Week 16	Full course
Assignments / Reading tasks	During semester	Concepts, algorithms, and short reviews
Paper / Project presentation	Week 15	Recent FL paper or project discussion
Participation	Every week	Discussion and class engagement

Suggested percentage allocation

Midterm 25% · Final 25% · Assignments/reading 20% · Paper/project presentation 20% · Participation 10%

Expected Background

What students should know before going deeper

Machine learning basics

Supervised learning, loss functions, training and testing, overfitting, evaluation metrics.

Optimization basics

Gradient descent, stochastic gradient descent, learning rate, convergence intuition.

Programming skills

Python is useful. PyTorch or TensorFlow experience is helpful, but not required at the beginning.

Mathematical comfort

Vectors, matrices, basic probability, and reading algorithmic notation.

Research reading

Students should be ready to read survey papers and recent FL research papers critically.

Week 1 Learning Objectives

By the end of this session, students should be able to

Explain why federated learning is needed in modern data-driven systems.

Distinguish federated learning from centralized, local-only, and distributed learning.

Describe the basic FL workflow using clients, server, local training, aggregation, and communication rounds.

Identify cross-device and cross-silo FL settings.

Recognize major FL application domains and current research challenges.

Guiding question for today

When is federated learning more appropriate than centralized learning?

Why Federated Learning?

The motivation begins with data that cannot easily be centralized

Privacy and regulation

Sensitive data may be protected by privacy laws, institutional rules, or user expectations.

Data ownership

Hospitals, companies, and users may not want to reveal raw data to a central owner.

Distributed data generation

Data is naturally generated on phones, sensors, vehicles, hospitals, and edge devices.

Communication and storage cost

Moving all raw data to one location can be costly, slow, or infeasible.

Collaboration without raw data sharing

Multiple parties can benefit from a shared model while retaining local control over data.

The Centralized Learning Assumption

Traditional ML usually assumes all training data can be collected in one place

Centralized ML

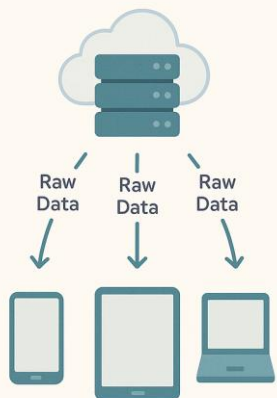
All raw data is uploaded to a central repository. The model is trained in one controlled environment.

Challenge

But in many real systems...

Data is private, distributed, legally constrained, large, noisy, imbalanced, and owned by different parties.

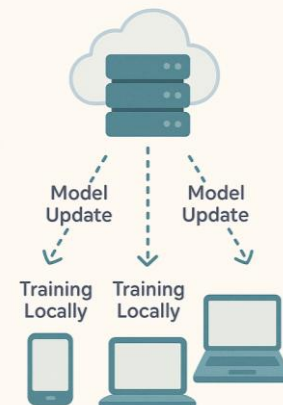
CENTRALIZED LEARNING



Main issue

Centralizing data may create privacy risk, regulatory risk, trust issues, and large communication/storage overhead.

FEDERATED LEARNING



What is Federated Learning?

A basic definition for this course

Federated learning is a machine learning setting where multiple clients collaboratively train a model while keeping raw training data local.

What moves?

Model updates, gradients, parameters, or other focused information.

What stays local?

Raw data remains on the client, device, organization, or silo.

What coordinates?

A server or aggregator orchestrates rounds and combines updates.

Definition aligned with Kairouz et al. (2021): clients collaborate under a central server while keeping training data decentralized.

Four Learning Settings

A quick comparison before studying FL algorithms

Centralized

Raw data moves to one place; one model is trained centrally.

Local-only

Each client trains alone; no collaboration or shared global knowledge.

Distributed

Data is centrally managed and split across compute nodes.

Federated

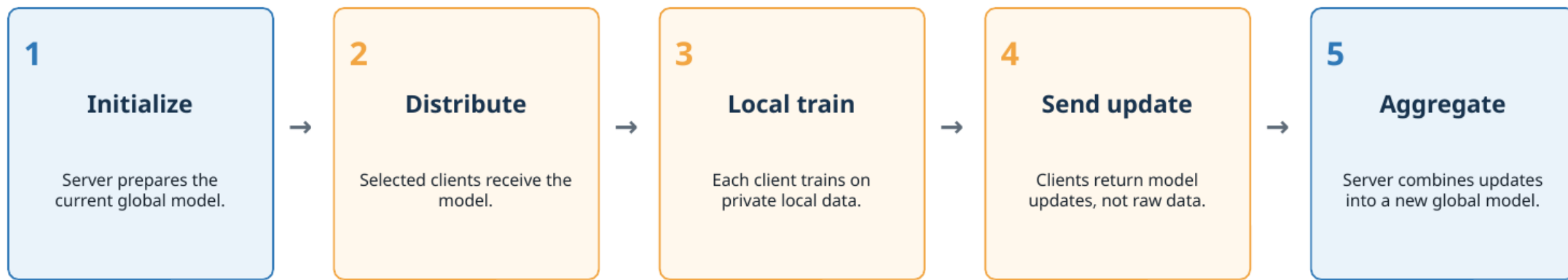
Data stays local; clients send focused updates for aggregation.

Key distinction

Federated learning is not simply distributed learning: data ownership, privacy, non-IID data, and unreliable clients matter.

Basic Federated Learning Workflow

A clean view of one federated training cycle



Round-based training

This process repeats for many communication rounds until the model converges or a stopping condition is met.

Core idea: move model information, keep raw data local

Key Terminology

Words we will use throughout the semester

Client / Party / Device

An entity that owns local data and participates in training.

Server / Aggregator

The coordinator that sends model information and aggregates updates.

Local Model

The model updated on a client using private local data.

Global Model

The shared model produced after aggregation.

Communication Round

One cycle of sending a model, local training, and aggregation.

Model Update

The information sent from client to server, such as weights or gradients.

Cross-Device vs Cross-Silo FL

Two common deployment settings

Cross-Device FL

Clients are very large numbers of mobile or IoT devices.

Typical traits: unreliable availability, limited bandwidth, partial participation, and no direct control over every device.

Cross-Silo FL

Clients are organizations, hospitals, companies, or data centers.

Typical traits: fewer clients, more stable participation, stronger identity, and institutional data ownership.

Why it matters

Algorithm design, privacy requirements, systems constraints, and evaluation strategy can differ between settings.

Major Application Domains

Where federated learning is useful

Healthcare

Mobile devices

IoT and edge computing

Smart homes

Finance

Recommender systems

NLP

LLM adaptation

Common pattern

The data is valuable for model training, but direct sharing of raw data is difficult because of privacy, ownership, regulation, cost, or deployment constraints.

Graduate-level question

For each domain, what is the actual reason FL is needed? Privacy, cost, regulation, personalization, or scalability?

Important Clarification

Federated learning helps privacy, but does not automatically solve it

What FL reduces

The need to collect raw data in a central repository.

What can still leak

Model updates, gradients, intermediate models, and final models may reveal information.

What may be needed

Differential privacy, secure aggregation, cryptography, trusted execution, or robust protocols.

Key message

“Data stays local” is not the same as “privacy is fully guaranteed.” Privacy analysis depends on the threat model.

Current Trends and Open Challenges

These topics will structure the rest of the course

Non-IID data

Clients have different data distributions.

Communication efficiency

Repeated model updates can be expensive.

Personalization

One global model may not fit all clients.

Privacy and security

Updates can leak information or be attacked.

Scalability

FL systems must handle many clients and failures.

FL for LLMs

Large models need parameter-efficient and resource-aware adaptation.

Discussion and Takeaways

End of Week 1

In-class discussion

Choose one scenario: healthcare, mobile keyboard, smart home, finance, recommender system, or LLM adaptation.

Discuss: Why might FL be better than centralized learning? What could still go wrong?

Week 1 takeaways

FL enables collaborative model training without sharing raw data.

The standard workflow involves clients, a server, local training, aggregation, and repeated rounds.

FL is motivated by privacy, ownership, regulation, and distributed data generation.

Before Week 2

Review basic optimization concepts: empirical risk minimization, loss functions, gradient descent, stochastic gradient descent, and convergence intuition.

Q & A

Thank You.

sofar@sejong.ac.kr